

Caitlin L. Le

📍 Princeton, NJ ✉ caitlin.le@princeton.edu in caitlin-le

EDUCATION

- PhD** **Princeton University**, Mechanical Engineering and Materials Science July 2026 – Present
- Advisor: Rebecca Kramer-Bottiglio
- MS** **Yale University**, Mechanical Engineering and Materials Science Aug 2023 – June 2026
- Advisor: Rebecca Kramer-Bottiglio
- BS** **University of California San Diego**, Mechanical Engineering, specialization in Controls and Robotics Sept 2019 – June 2023
- GPA: 3.9/4.0
 - Cum Laude

RESEARCH EXPERIENCE

- The Faboratory**, *Yale and Princeton University* New Haven, CT
PhD Candidate, Advisor: Rebecca Kramer-Bottiglio Aug 2023 – Present
- Investigate how soft robotic mechanisms influence burrowing efficiency, integrating concepts from soft robotics and geotechnical engineering.
 - Develop bioinspired alternatives to fluid jetting for subsurface penetration; funded by a NASA Space Technology Graduate Research Opportunities Award (80NSSC24K1339).
 - Manage the lab's science communication efforts across social media platforms, creating multimedia content that translates complex research into accessible public-facing material.
- NASA Glenn Research Center** Cleveland, OH
Visiting Technologist Experience May 2026 - June 2026
- As part of the NASA Space Technology Graduate Research Opportunities Award, conducted research in the Excavation Lab on the effects of intruder softness on intrusion force in GRC-3b simulant.
 - Characterized the resetting consistency of the Excavation Lab's granular testbed setup for loose and compact GRC-3b simulant.
- NASA Glenn Research Center** Cleveland, OH
Visiting Technologist Experience May 2025 - July 2025
- As part of the NASA Space Technology Graduate Research Opportunities Award, conducted research in the TREC Lab on the effects of intruder softness on intrusion force in GRC-3 simulant.
- Gravish Lab**, *University of California San Diego* La Jolla, CA
Undergraduate Researcher, Advisors: Nick Gravish and Michael T. Tolley Sept 2020 – June 2023
- Studied bioinspired locomotion in granular media, analyzing cyclic pumping and fluidization frequencies to reduce drag forces during subsurface motion in submerged granular media.
 - Designed and experimentally tested compliant anisotropic appendages to investigate how spacing affects intruder propulsion through dry granular substrates.

PUBLICATIONS

Journal Publications

- [J1] **C. L. Le[†]**, O. D. Yirmibesoglu[†], S. Even, T. Buckner, Y. Ozkan-Aydin, and R. Kramer-Bottiglio. (2025). Grand challenges for burrowing soft robots. *Frontiers in Robotics and AI*, vol. 12, pp. 1525186. [†]*These authors contributed equally.* Accompanying video: <https://youtu.be/AtW663xQLug?si=m12xNvnkX1HSTW2b> 

Conference Publications

- [C1] **C. L. Le**, F. Fonseca Cagnazzo[†], N. Ramos[†], E. I. Figueroa, C. Creager, and R. Kramer-Bottiglio. (2026). Common Ground: Standardizing Granular Media Characteristics for Burrowing Robots. *9th IEEE International Conference on Soft Robotics*. [†]*These authors contributed equally.*
- [C2] M. S. Li, B. H. Do, **C. L. Le**, C. S. O'Hern, and R. Kramer-Bottiglio. (2025). Variable Stiffness and Variable Size Particles for Reconfigurable Granular Metamaterials. *8th IEEE International Conference on Soft Robotics*.

Posters

- [P1] "Common Ground: Standardizing Granular Media Characteristics for Burrowing Robots" poster at 1st Princeton Robotics Symposium, Princeton, New Jersey, April 2026.
- [P2] "Common Ground: Standardizing Granular Media Characteristics for Burrowing Robots" poster at 9th IEEE International Conference on Soft Robotics, Kanazawa, Japan, April 2026.
- [P3] "Effects of Intruder Modulus on Burrowing Forces in Granular Media," poster at *Machine-Ground Interaction Consortium*, Madison, Wisconsin, September 10, 2025.
- [P4] "Effects of Softness on Burrowing Forces in Granular Media," poster at *8th IEEE International Conference on Soft Robotics*, Lausanne, Switzerland, April 25, 2025.
- [P5] "Frequency of Cyclic Water Pumping as a Drag Reduction Mechanism for Robot Locomotion in Submerged Granular Media," poster at *University of California San Diego Guided Engineering Apprenticeship in Research Symposium*, May 2021. Accompanying video: <https://youtu.be/hMeOdi8wPQI?si=sVpLeB0l0lGwn50a> 

TEACHING EXPERIENCE

Mechanical Design (MENG 185, 1105), Yale University

Teaching Fellow with Prof. Rebecca Kramer-Bottiglio and Prof. Tamanna Rahman

Fall 2024, Spring 2025,
Spring 2026

- Led laboratory sessions and guided students through hands-on projects in robot design, fabrication, and programming.
- Provided technical mentorship and feedback on mechanical design fundamentals, prototyping, and system integration.

AWARDS

NASA Space Technology Graduate Research Opportunities (NSTGRO 2024)

2024

Proposal title: Resource Ingesting Soft Robotic skins for In Situ Regolith Sampling (80NSSC24K1339)

Triton Research & Experimental Learning Scholars (TRELS) Quarterly Award

2021

University of California San Diego

PROFESSIONAL EXPERIENCE

IDEA Engineering Student Center, University of California San Diego

Lead Peer Educator

La Jolla, CA
Jan 2020 – June 2023

- Taught physics and chemistry coursework to undergraduate engineering students in one-on-one and group settings.

Collins Aerospace

Research and Technology Intern

Chula Vista, CA
June 2022 – August 2022

- Designed and analyzed a foldable wingtip concept for commercial aircraft, integrating morphing structures capable of both taxiing configurations and gust-load alleviation in flight; submitted two IP disclosure (pending patents).
- Performed aerodynamic simulations in STAR-CCM+ and developed CAD models in CATIA.
- Presented design concepts and performance results to industry partners at UC San Diego's Corporate Affiliates Program executive board meeting.

OUTREACH AND VOLUNTEER SERVICE

Social Assistance Program for Vietnam, Gia Lai, Vietnam

July 2026

Dental Education Assistant

- Assembled and distributed 3,000+ dental hygiene kits to underserved communities with limited access to dental care.
- Taught proper toothbrushing technique and preventive oral hygiene practices to 3,000+ residents, many with no prior dental education.

Graduate Engineering Community Organization, Yale University

Aug 2024 – May 2026

Project Manager

- Organized community-building events for engineering graduate students, helping create a more connected and supportive academic environment.

Silliman College Graduate Affiliate, Yale University

Sept 2023 – May 2026

- Facilitated social events that strengthened graduate-undergraduate interaction.

Yale Graduate Society of Women Engineers, Yale University

Nov 2024 – Dec 2025

Graduate Mentor

- Mentored undergraduate women in engineering, offering guidance on research pathways, graduate school preparation, and navigating academic environments.

Graduate Women in SEAS Mentorship Program, Yale University

Aug 2024 – May 2025

Organizer

- Coordinated mentoring events involving graduate students and faculty, fostering community and strengthening support networks for women in STEM.

Girls' Science Investigations 2024, Yale Pathways

Sept 2024

Volunteer

- Taught middle school students about rocket propulsion through hands-on demonstrations, promoting early interest and engagement in engineering.

Good Samaritans Medical and Dental Ministry, Can Tho, Vietnam

July 2023

Lab Assistant

- Conducted blood, lipid, and urinalysis panels for 500+ patients in Can Tho, Vietnam, supporting medical outreach in underserved communities.

EXTRACURRICULAR ACTIVITIES

Triton Robotics, University of California San Diego

Sept 2019 – June 2023

Mechanical Subteam Lead, Senior Advisor

- Led the mechanical subteam in designing a RoboMaster competition robot, overseeing mechanical development, actuator selection, and subsystem integration.
- Developed and analyzed CAD models for the robot's claw mechanism to optimize performance during competition tasks.
- Mentored junior members and served on the board, representing the organization at STEM outreach events.

NASA L'Space Academy

May 2021 – July 2021

Principal Investigator

- Led a student team in designing a burrowing robot for lunar subsurface exploration, authoring a full proposal rooted in soft bioinspired robotics.
- Directed technical development, coordinated research tasks, and submitted a New Technology Report to NASA detailing the design's potential applications.